CSC1042 Computer Network II

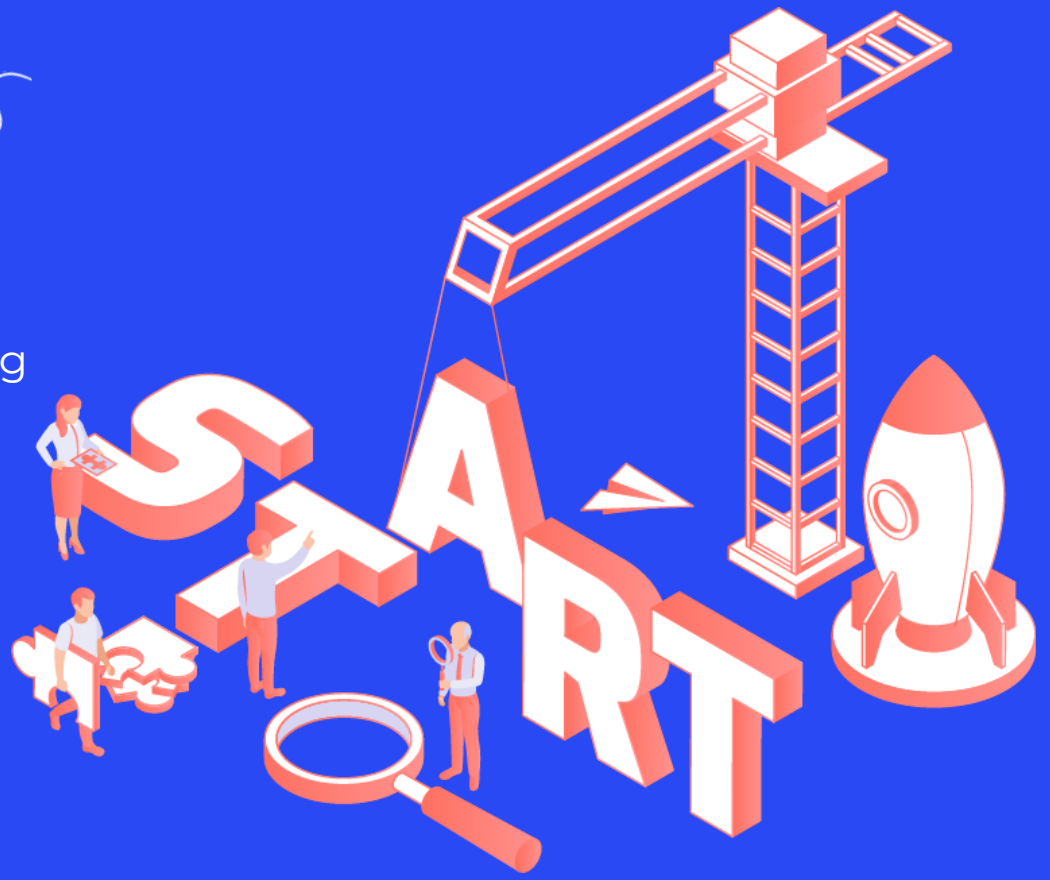
Distance Vector Routing

Dr. Mohammed Amine TOGOU



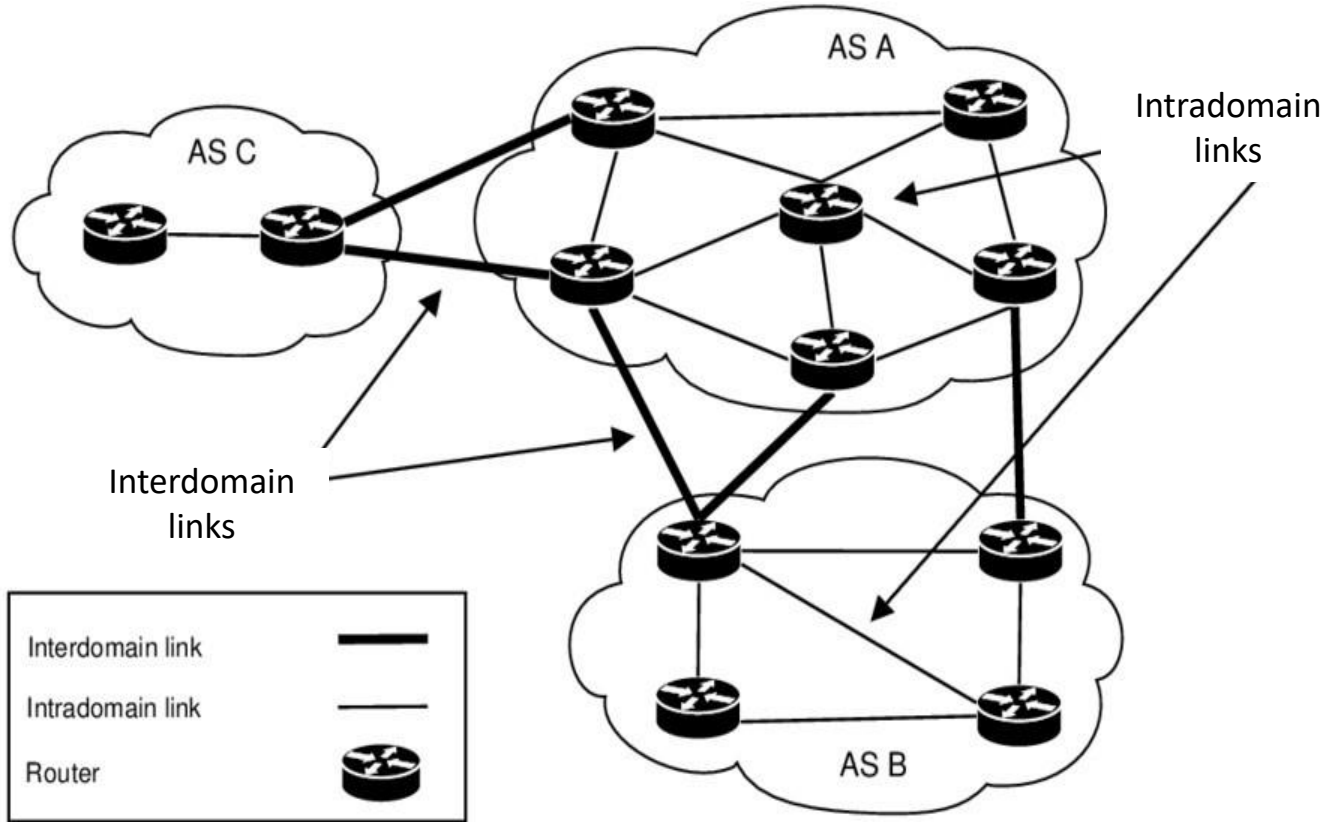
Learning outcomes

- Explain the concept of distance vector routing
- Describe how routers build and update their routing tables using distance vector routing
- Discuss the challenges of distance vector routing protocols and their solutions



Autonomous system (AS)

- A network or a group of networks that are all managed and supervised by a single entity or organisation
- e.g., DCU's networks, IBM networks, amazon networks, etc.



Intradomain

- It focuses on routing within AS
- It ignores the Internet outside AS
- Protocols for Intradomain routing are also called Interior Gateway Protocols (IGPs)
- Popular protocols:
  - Routing Information Protocol (RIP) – old but simple
  - Open Shortest Path First (OSPF) – better performance
  - Enhanced Interior Gateway Routing Protocol (EIGRP)

Interdomain

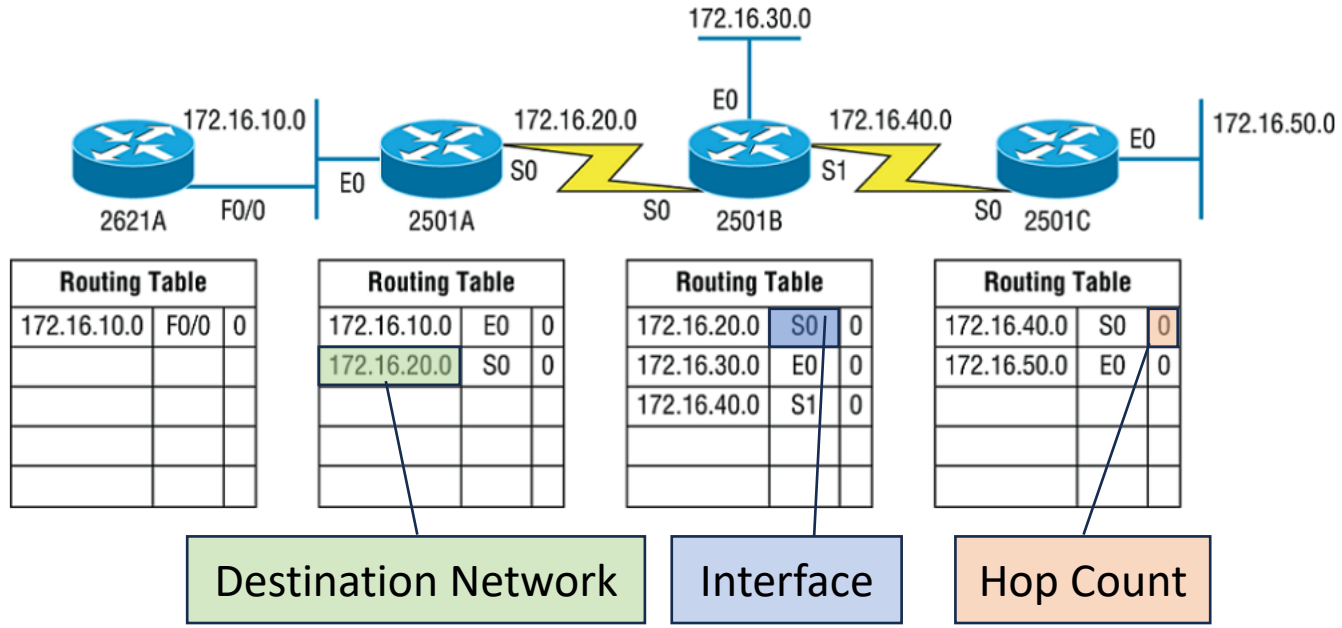
- It focuses on routing between autonomous systems
- It assumes that Internet consists of a collection of interconnected autonomous systems
- Usually, there is one dedicated router in each AS handling interdomain traffic
- Interdomain protocols are called Exterior Gateway Protocols (EGPs)
  - e.g., Border Gateway Protocol (BGP) – this is recent

Distance vector routing

- Distance means how far
  - Can be measured in terms of number of hops, delay, etc.
- Vector means in what direction
- It is an old and very simple routing approach
- Each router knows the distance to its directly connected neighbors
- The routers update each other on a fixed interval
- It uses hop count to determine the best path to remote networks
- Very good for small and simple networks as it can be easily implemented

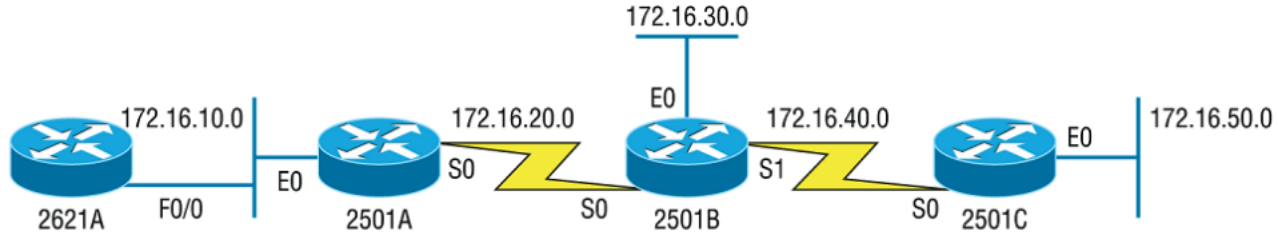
Distance vector routing – How it works?

- Let's look at this example
- At the start, every router only has the directly connected networks in its routing table
  - Note the hop count is 0 in all cases
- These routing tables will be exchanged between the 4 routers after some time



Distance vector routing – How it works?

- Every router will then update its routing table using Bellman-Ford algorithm based on the routing tables received from other routers



Routing Table		
172.16.10.0	F0/0	0
172.16.20.0	F0/0	1
172.16.30.0	F0/0	2
172.16.40.0	F0/0	2
172.16.50.0	F0/0	3

Routing Table		
172.16.10.0	E0	0
172.16.20.0	S0	0
172.16.30.0	S0	1
172.16.40.0	S0	1
172.16.50.0	S0	2

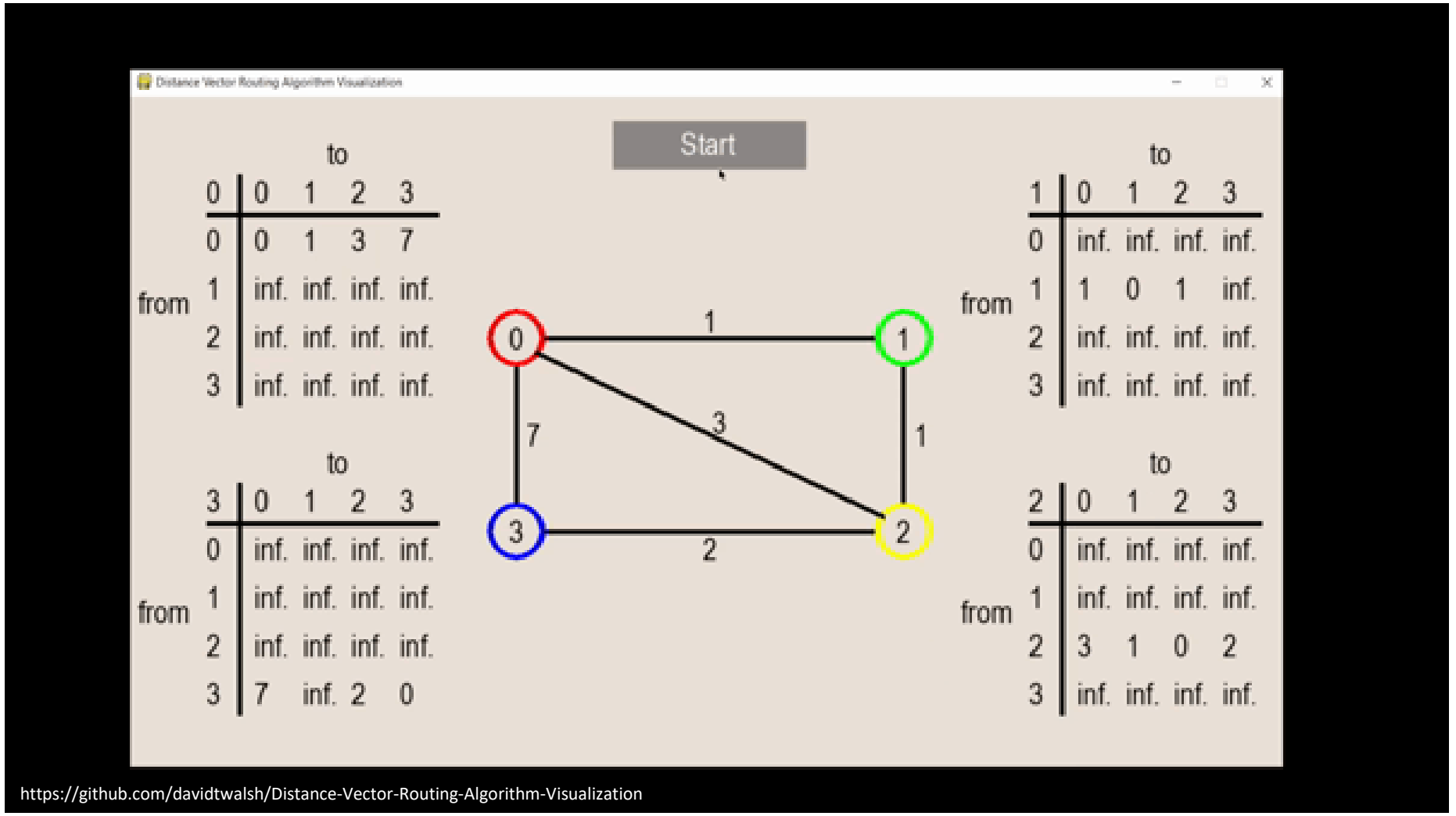
Routing Table		
172.16.20.0	S0	0
172.16.30.0	E0	0
172.16.40.0	S1	0
172.16.10.0	S0	1
172.16.50.0	S1	1

Routing Table		
172.16.40.0	S0	0
172.16.50.0	E0	0
172.16.10.0	S0	2
172.16.20.0	S0	1
172.16.30.0	S0	1

- Note that while the hop count for directly connected networks remain 0, the hop count for remote networks is incremented by 1 each time the path completely passes through a router

- Note that while routers are building their routing table, no data transmission will take place. Only when all routers converge (i.e., their routing table are completed), then data transmission can start

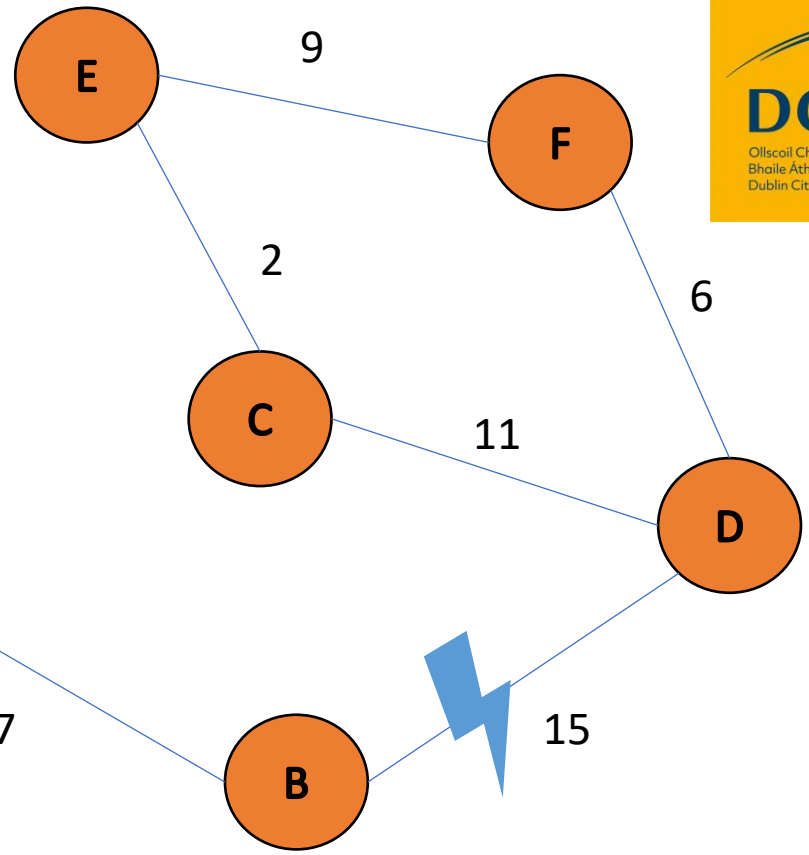
Distance vector routing



https://github.com/davidtwalsh/Distance-Vector-Routing-Algorithm-Visualization

Distance Vector Routing – Count to infinity

- A fatal flow in distance vector routing
- Can occur when a link/router goes down
- Assume the link between B and D goes down

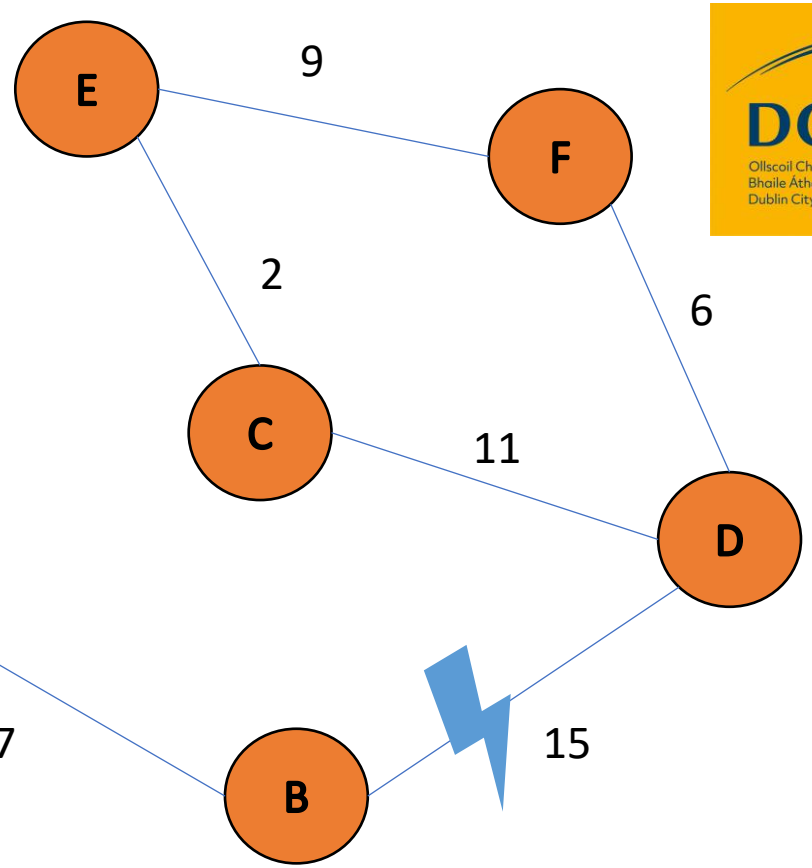


To	Distance
B	7
C	33
D	22
E	35
F	28

To	Distance
A	7
C	26
D	15
E	28
F	21

Distance Vector Routing – Count to infinity

- Assume the link between B and D goes down
- B now has no route to D

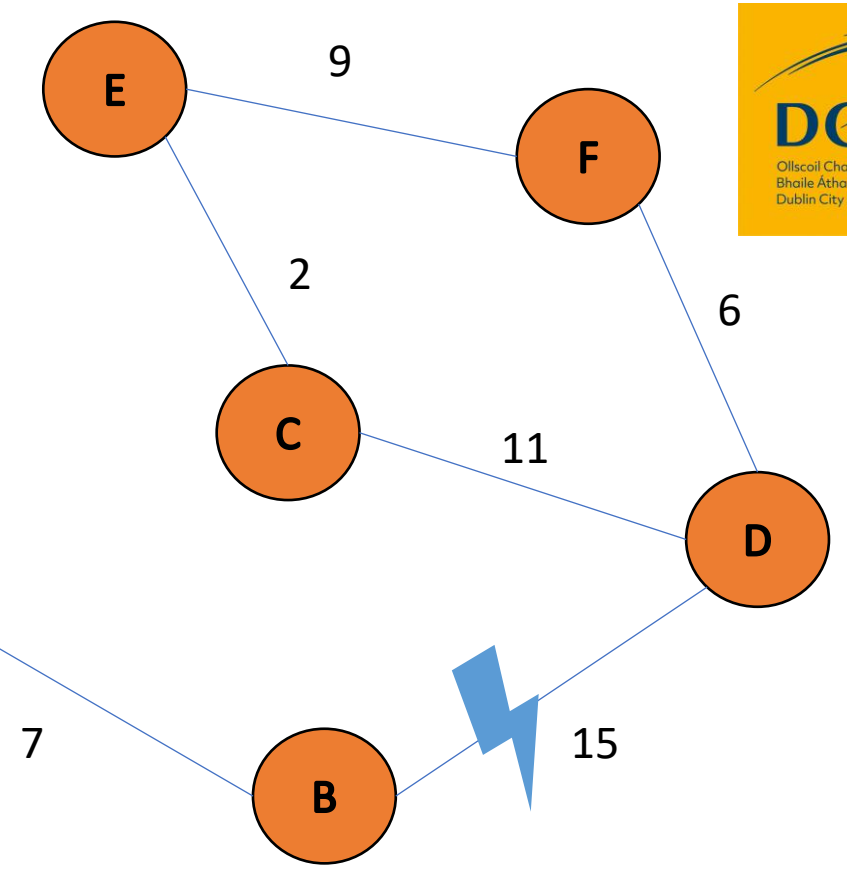


To	Distance
B	7
C	33
D	22
E	35
F	28

To	Distance
A	7
C	26
D	-
E	28
F	21

Distance Vector Routing – Count to infinity

- Assume the link between B and D goes down
- B now has no route to D
- Before B can announce that D is unreachable, A sends its periodic routing table
- B update its routing table accordingly
- B thinks that there is a path to D through A

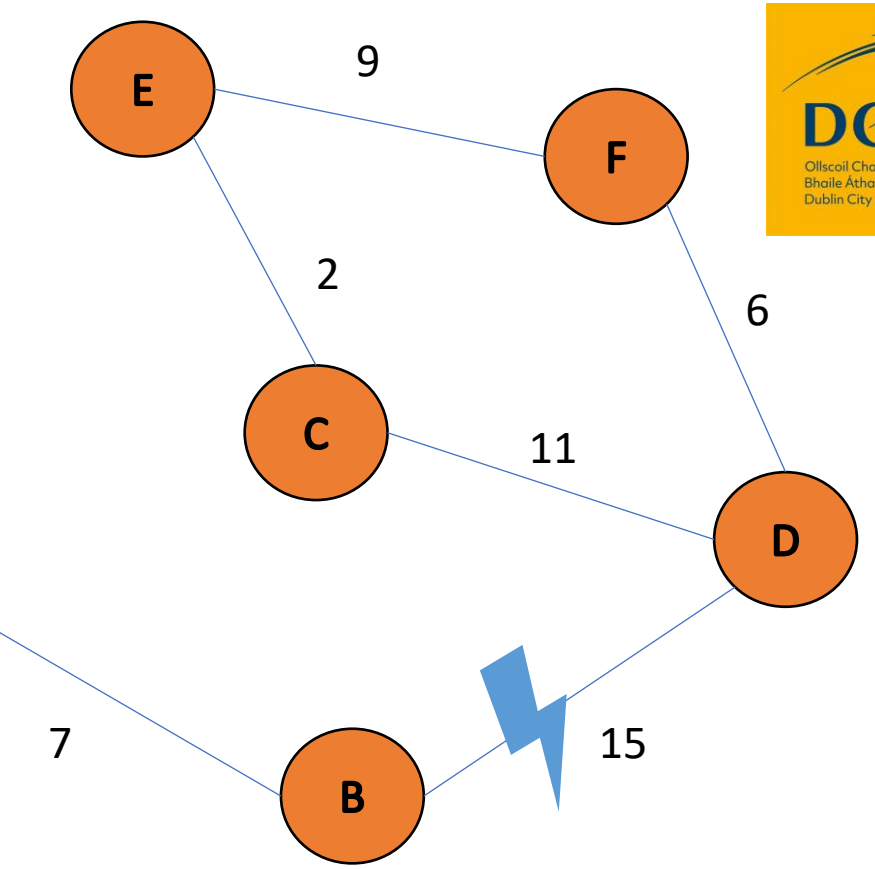


To	Distance
B	7
C	33
D	22
E	35
F	28

To	Distance
A	7
C	26
D	29
E	28
F	21

Distance Vector Routing – Count to infinity

- B sends its routing table to A
- A still thinks that there is a path to D through B
- A updates its routing table accordingly
- This process continues indefinitely



To	Distance
B	7
C	33
D	36
E	35
F	28

To	Distance
A	7
C	26
D	29
E	28
F	21

Routing Information Protocol (RIP)

- A distance vector routing protocol
  - An application layer protocol using the UDP port 520
- Using RIP, every router sends its complete routing table to its neighboring routers every 30 seconds
- It uses hop count to determine the best path to a remote network
  - Default maximum hop count is 15 – used to avoid loops
- It works well in small networks, but it's inefficient in large networks due its slow convergence time

Routing Information Protocol (RIP)

- It has 2 versions
  - RIPv1
    - Only uses classful routing
    - Uses broadcast for route updates
    - No verification of the source – All broadcasts are accepted

      → big security issue
  - RIPv2
    - Supports VLSM by sending subnet mask info with the routing table
    - Uses multicast for route updates
    - Supports router authentication

Routing Information Protocol (RIP) - challenges

- Invalid routes can stay in a routing table for several minutes
  - Routers end up sending traffic to “black holes”
  - Solution: if a link goes down, send an immediate update
- Count to infinity problem
  - Solutions:
    - Max hop count of 15
    - Split horizon: a router does not advertise routes to the router from which it learned those routes

THANK YOU